

COURSE PORTFOLIO

Natural Language Processing

Nichelle Graf *AI Focus: Bridging the gap between humans and machine*

Applied Artificial Intelligence & Robotics
Houston City College
ITAI 2373 – NLP | Spring 2026
Professor Anna Devarakonda

<https://github.com/NichelleG/NLP-ITAI2373>



Learning Journey

Foundation: Trade Tools, Toolkits & Text Processing

GitHub · Jupyter · VS Code · Python · NLTK · spaCy · Pandas

Core NLP: Text Representation / POS / Semantic Analysis

BoW · TF-IDF · N-Grams · Scikit-Learn · Matplotlib

Machine Learning: Sentiment & Emotion / Text Classification

VADER · TextBlob · DataFrames · Seaborn

Advanced Techniques: LLMs / Machine Translation / Topic Modeling

LDA · NMF · NER · Conversational AI

Final Project: NewsBot Intelligence System 2.0

Kaggle · Production-Ready Platform · Multi-Pipeline NLP

<https://github.com/NichelleG/NLP-ITA12373>

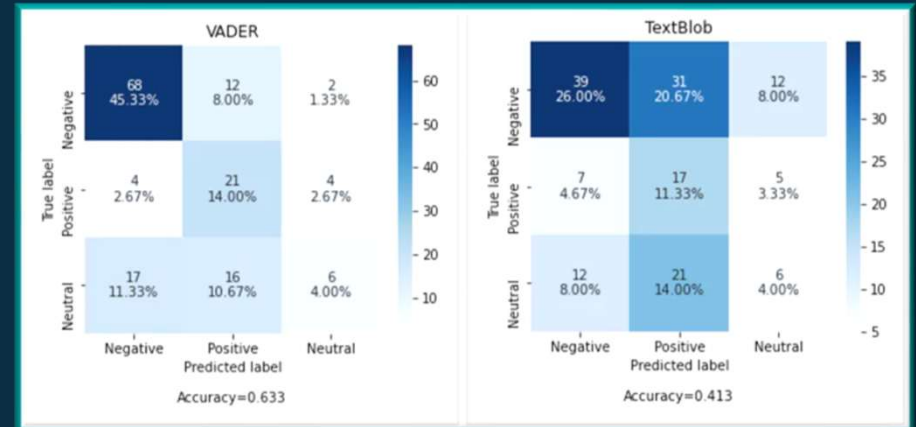
Featured Projects

1

Sentiment & Emotion Analysis Lab

Multi-library pipeline comparing **VADER** and **TextBlob** across social media, reviews, and news datasets.

Pandas · Seaborn · DataFrames



2

NewsBot Intelligence System (Development) Midterm Project



3

NewsBot 2.0 Enhancement (Production Platform) Final Project



Advanced News Content Analysis:
Topic modelling > Multi-class classification >
Querying > Interpretable outputs.

TF-IDF · LDA · NMF · NER · Kaggle

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Skills & Competencies

NLP CORE	TEXT REPRESENTATION	ML PIPELINE
• Tokenization & POS Tagging • Named Entity Recognition (NER) • Stemming & Lemmatization • Sentiment & Emotion Analysis	• Bag of Words (BoW) • TF-IDF Vectorization • N-Grams & Word Embeddings • Topic Modeling: LDA / NMF	• Text Classification • Train/Test Evaluation • Preprocessing Pipelines • Bias & Ethics Awareness

TOOLS & LIBRARIES USED THIS SEMESTER

Python



NLTK



spaCy



Pandas



Scikit-Learn



VADER



TextBlob



Seaborn



Jupyter



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Reflection & Next Steps

KEY INSIGHTS

- Preprocessing decisions are as critical as model choice — the same technique can help or hurt depending on context.
- Word embeddings changed how I think about meaning: semantic relationships are directions in vector space, not labels.
- Every NLP pipeline involves trade-offs between speed, accuracy, and information loss — there is no universal solution.
- Ethics isn't a footnote — bias encoded in training data becomes bias encoded in decisions.

NEXT STEPS

- 01** Explore transformer architectures (BERT, GPT) and fine-tuning for domain-specific NLP tasks.
- 02** Integrate NewsBot 2.0 with live news APIs for real-time querying and summarization.
- 03** Apply NLP skills to cybersecurity use cases — log analysis, threat intelligence extraction.
- 04** Publish annotated Jupyter notebooks to GitHub as a public learning resource for other students.

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